

Carleton University

Course Project: SCARA Robot Report

MECH 4503A: An Introduction to Robotics

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1 Introduction

2 SCARA Robot Architecture

2.1 Mechanical Configuration

The canonical 4-DOF SCARA robot (Figure 1) consists of a fixed base, two horizontal revolute links of lengths a_1 and a_2 , a vertical prismatic joint, and a final revolute wrist joint. Table 1 summarises the joint types and their physical roles [1][2].

Joint	Type	Symbol	Motion	Role
1	Revolute	θ_1	Rotation about Z_0	Shoulder rotation in XY plane
2	Revolute	θ_2	Rotation about Z_1	Elbow rotation in XY plane
3	Prismatic	d_3	Translation along Z_2	Vertical (Z) positioning
4	Revolute	θ_4	Rotation about Z_3	End-effector spin / tool orientation

Table 1: SCARA joint summary [1][2].

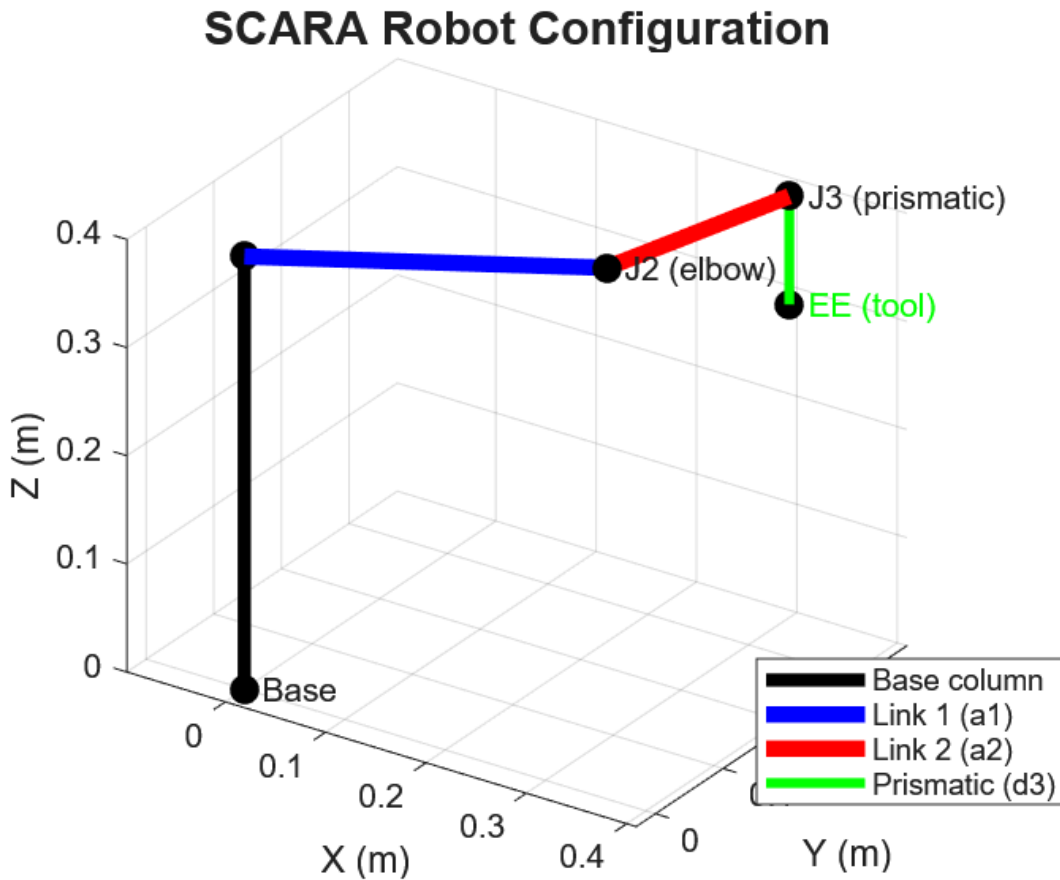


Figure 1: Typical 4-DOF SCARA robot configuration